

# CS5288: CRYPTOGRAPHY: THEORY AND PRACTICE

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## Effective Term

Semester A 2025/26

## Part I Course Overview

### Course Title

Cryptography: Theory and Practice

### Subject Code

CS - Computer Science

### Course Number

5288

### Academic Unit

Computer Science (CS)

### College/School

College of Computing (CC)

### Course Duration

One Semester

### Credit Units

3

### Level

P5, P6 - Postgraduate Degree

### Medium of Instruction

English

### Medium of Assessment

English

### Prerequisites

(CS5222 Computer Networks and Internets or equivalent) and  
(MA2144 Discrete Mathematics or equivalent)

### Precursors

Nil

### Equivalent Courses

Nil

### Exclusive Courses

Nil

## Part II Course Details

### Abstract

The course provides an in-depth study of cryptographic techniques and their role in practical computer systems and applications. It covers the algorithms for symmetric and asymmetric encryption, hash functions, and pseudo random number generation; and the protocols to achieve practical security objectives such as confidentiality, authentication, data integrity, non-repudiation. Associated protocols such as key distribution and public key infrastructure systems will also be dealt with.

### Course Intended Learning Outcomes (CILOs)

| CILOs |                                                                                                                                           | Weighting (if app.) | DEC-A1 | DEC-A2 | DEC-A3 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|--------|--------|
| 1     | Apply modular arithmetic mathematic and basic group theoretic/finite field operations related to cryptographic techniques.                |                     |        | x      |        |
| 2     | Describe basic concepts and algorithms of cryptography, including encryption/decryption, hash functions, pseudo random number generation. |                     |        | x      |        |
| 3     | Critique and assess the security of cryptographic functions, and evaluate their strength.                                                 |                     |        | x      |        |
| 4     | Create and analyze protocols for various security objectives with cryptographic tools.                                                    |                     | x      | x      | x      |
| 5     | Explain the impact of potential future development of cryptography such as quantum cryptography.                                          |                     | x      | x      | x      |

#### A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

#### A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

#### A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

### Learning and Teaching Activities (LTAs)

| LTAs |           | Brief Description                                                          | CILO No.   | Hours/week (if applicable) |
|------|-----------|----------------------------------------------------------------------------|------------|----------------------------|
| 1    | Lecture   | Students will engage in lectures about concepts, theory and methodologies. | 1, 2, 3, 5 | 2 hours/ week              |
| 2    | Tutorials | Students will engage in exercises and discussions                          | 1, 2, 4, 5 | 1 hour/ week               |

### Assessment Tasks / Activities (ATs)

|   | ATs           | CILO No. | Weighting (%) | Remarks ("- for nil entry) | Allow Use of GenAI? |
|---|---------------|----------|---------------|----------------------------|---------------------|
| 1 | Assignment #1 | 1, 2, 4  | 10            | -                          | Yes                 |
| 2 | Assignment #2 | 1, 2     | 10            | -                          | Yes                 |
| 3 | Quiz          | 1, 2, 5  | 10            | -                          | No                  |

**Continuous Assessment (%)**

30

**Examination (%)**

70

**Examination Duration (Hours)**

2

**Minimum Examination Passing Requirement (%)**

30

**Additional Information for ATs**

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

**Assessment Rubrics (AR)****Assessment Task**

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

Ability to explain and use concepts, algorithms and protocols, and the ability to solve problems independently

**Excellent**

(A+, A, A-) High

**Good**

(B+, B, B-) Significant

**Fair**

(C+, C, C-) Moderate

**Marginal**

(D) Basic

**Failure**

(F) Not even reaching marginal levels

**Assessment Task**

Quiz (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

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**Assessment Task**

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

**Criterion**

Ability to explain and use concepts, algorithms and protocols, and the ability to solve problems independently

**Excellent**

(A+, A, A-) The answer is correct and written in a clear manner.

**Good**

(B+, B) The answer is mostly correct, with some minor mistakes.

**Marginal**

(B-, C+, C) The answer is large correct with some mistakes.

**Failure**

(F) Below the marginal level

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### Assessment Task

Quiz (for students admitted from Semester A 2022/23 to Summer Term 2024)

#### Criterion

Ability to explain and use concepts, algorithms and protocols

#### Excellent

(A+, A, A-) The answer is correct and written in a clear manner.

#### Good

(B+, B) The answer is mostly correct, with some minor mistakes.

#### Marginal

(B-, C+, C) The answer is large correct with some mistakes.

#### Failure

(F) Below the marginal level

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### Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

#### Criterion

Ability to explain and use concepts, algorithms and protocols

#### Excellent

(A+, A, A-) Depending on the rubrics of the final exam paper

#### Good

(B+, B) Depending on the rubrics of the final exam paper

#### Marginal

(B-, C+, C) Depending on the rubrics of the final exam paper

#### Failure

(F) Depending on the rubrics of the final exam paper

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## Part III Other Information

### Keyword Syllabus

Basic number theory, one-way functions, basic randomness, symmetric encryption, one-time Pad, Feistel structure, DES, IDEA, AES, brute force attacks, strength of encryption functions, block and stream cipher, key distribution problem, secret sharing, asymmetric encryption, RSA, prime number generation, public key protocol, hybrid encryption, key exchange protocol, Diffie-Hellman, authentication protocols, hash functions, MD5, SHA, data integrity, message integrity code, non-repudiation, digital signature, RSA signature, ElGamal, DSA, elliptic curve cryptosystem, trust model, digital certificate, PKI, zero knowledge proofs, blind signature, quantum cryptography.

### Reading List

#### Compulsory Readings

| Title |                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Handbook of Applied Cryptography Online version: <a href="http://www.cacr.math.uwaterloo.ca/hac/">http://www.cacr.math.uwaterloo.ca/hac/</a> |
| 2     | William Stallings, Cryptography and Network Security: Principles and Practices. Prentice Hall, ISBN-10: 0136097049, 5th edition.             |

**Additional Readings**

| Title |                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Chapman & Hall. Cryptography, Theory and Practice. CRC, ISBN 1584882069, 2nd edition.                                                     |
| 2     | Cryptography: An Introduction                                                                                                             |
| 3     | Online version: <a href="https://www.cs.umd.edu/~waa/414-F11/IntroToCrypto.pdf">https://www.cs.umd.edu/~waa/414-F11/IntroToCrypto.pdf</a> |